

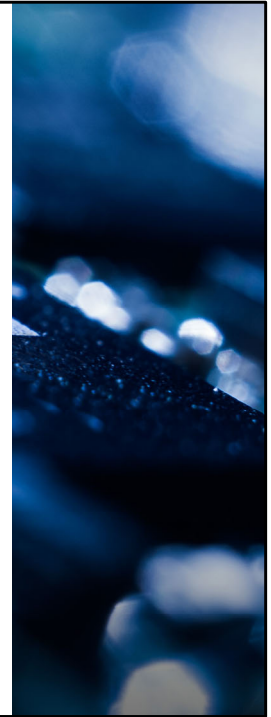


SC1006

Computer Organisation and Architecture

Data Centre Storage

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- This section is an introduction on **Data Centre Storage**

Criteria for Storage Element

- Power consumption
 - Data centers consumed a lot of power, which not only translates to direct costs in utility and cooling measures but also limits its choice of location.
 - Centres are commonly found near natural cooling elements such as large natural water bodies which offer a low-cost and reliable source of cooling.
- Speed
 - Beneficial for caching databases and other data affecting overall application or system performance.
- Robustness
 - Tolerance to various forms of mechanical movement/interference increases reliability and reduces the need and cost of maintenance.
 - housing structure shock absorption requirement is also reduced.
- Heat production
 - The less heat generated the less cooling and power required in the data centre.
- Size
 - Data centers will be able to store more data in less space: increasing efficiency in all areas (power, cooling, etc.)



This slide shows some of the key considerations when selecting storage elements to use in a Data Center.

First is the Power Consumption

Data Center uses many storage elements so power consumption of these storage elements is very important.

It translates to direct cost of electrical bill and also the indirect cost of requirement for cooling measures.

The need for cooling measures may also limit the choice of Data Center Locations, with many of the Centers located natural cooling sources such as natural water bodies.

Next is the speed factor

Having storage element with higher speed will improve the overall performance of the system hosted in the Data Centers. Data Centers these days are increasingly being used to host virtual machines and other processing related functions, rather than just a simple cloud storage.

The storage element needs to be robust as well too since reliability is one of the most

important attributes of a Data Center.

One of the desired attributes is tolerance of various forms of mechanical movement and interference. This will improve the reliability and reduces the need and cost for maintenance as the storage element is less likely to fail.

Cooling measures, as mentioned above, incur cost and limits the location choices. So having low power consumption will be very beneficial.

Lastly, would be good for the size to be small to be able to pack into a smaller building.

Criteria

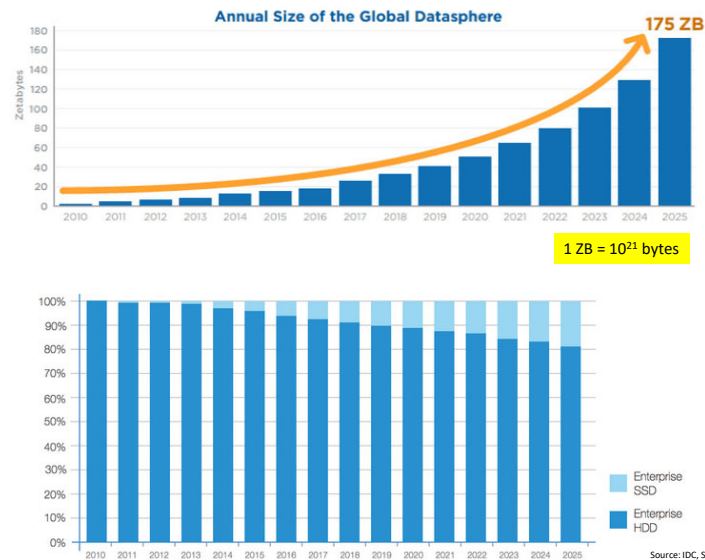
- Based on the criteria in the previous slide, SSD wins HDD hands down.
- So why is HDD still the dominant Storage in Data Centres?

COST



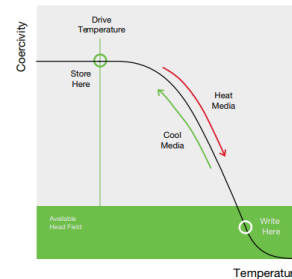
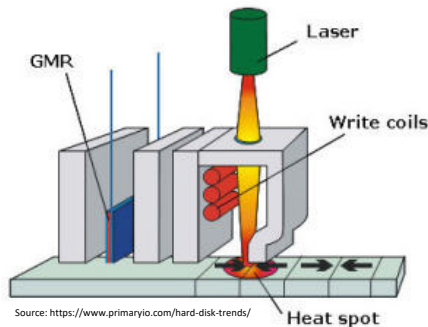
- Based on the selection criteria discussed in the previous slide, the obvious choice of storage element is the SSD.
 - But HDD is still the dominant storage for Data Center because of one most important reason – COST.
- The cost per bit of SSD today is still higher than that of HDD, this translate to huge cost difference due to the huge storage size of a Data Center.

Global Storage Consumption and Shipment Projection



- Other use cases that require huge storage also see HDD as the preferred choice, such as in the enterprise servers.
- As seen in the chart here, although SSD is slowly creeping into the market share of HDD, the total size of shipment for HDD is still much larger than that of SSD.

HAMR (Heat Assisted Magnetic Recording)



- The areal density of the HDD was stagnant for a while and manufacturers had been shipping drives with 2TByte/Platter.
- But with the HAMR, the areal density is expected to increase again.
- A small laser is attached to a recording head, designed to heat a tiny spot on the disk where the data will be written. This allows a smaller bit cell to be written as either a 0 or a 1.
- Current projections are that HAMR can achieve 5 Tbps, enabling hard drives with capacities higher than 100 TB.

Cost is the single most important reason why HDD is still surviving to this day, so naturally, the HDD maker is always looking at new technologies to increase their cost advantage.

One of these technologies is the HAMR, which stands for Heat Assisted Magnetic Recording.

There was a time where the area density of HDD was stagnant for a while and manufacturers had been shipping drives which has 2TB/platter.

But with HAMR, the areal density is expected to increase again.

The recording media used in HAMR drive allows more data to be packed into the same area reliably, but it is also more difficult to change the orientation of the magnetic element on these media.

So for HAMR, a smaller laser is attached to the recording head.

The laser is designed to heat a tiny spot on the disk, increasing the temperature allows data to be written more easily to the disk.

Current projection is that the HAMR drives can achieve 5Tbps, enabling drive with capacities higher than 100TB.

HAMR

- The major problem with packing bits so closely together on conventional magnetic media is that the data bits become unstable and may flip ($0 \rightarrow 1$ or $1 \rightarrow 0$).
- To make the media maintain their stability to store bits over a long period of time, the recording media needs to have a higher coercivity.
- Higher coercivity implies the media is magnetically more stable during storage, but it would also be more difficult to change the magnetic characteristics of the media when writing.
- For that, a laser is employed to heat a tiny region of several magnetic grains for a very short time (~ 1 ns) to a temperature high enough to lower the media's coercive field to below that of the write head's magnetic field. This is the write process.
- Immediately after the heat pulse, the region quickly cools down and the bit's magnetic orientation is frozen in place. The data is stored in the media and would be stable due to the high coercivity of the recording media.
- See the graph in the previous slide for the visual illustration.



This slides describe the principle for the HAMR drives.

One main problem of increasing the areal density of a HDD media is that the data bits will become unstable, meaning it has higher chance of flipping from 1 to 0 or vice versa.

To make the media more stable, the recoding media needs to have high coercivity. Higher coercivity means the magnetic media is more stable but it also means that it is more difficult to change its magnetic characteristics, i.e. more difficult to write to the media.

To counter that, a laser is used to heat up the spot where the head is writing to, this increases the temperature and lower the coercivity of that surrounding area. The reduced coercivity allow data write to be carried out properly.

After the write operation, the laser is turned off, the area cools down quickly and regain its coercivity and stability.

You can revisit the graph in the previous slide for illustration of this process.

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